Spacing in Math Mode

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June 28, 2026

Introduction

Assume we have the next sets

$$S = \{z \in \mathbb{C} \,|\, |z| < 1\} \quad \text{and} \quad S_2 = \partial S$$

Spaces

$\mathrm{f(x)} = \mathrm{x}^2 + 3x + 2$
$f(x) = x^2 + 3x + 2$
$f(x) = x^2 \, + 3x \, + 2$
$f(x) = x^2 \: + 3x \: + 2$
$f(x) = x^2 \; + 3x \; + 2$
$f(x) = x^2 \ + 3x \ + 2$
$f(x) = x^2 \quad + 3x \quad + 2$
$f(x) = x^2 \qquad + 3x \qquad + 2$

Operators Spacing

3ax+4by=5cz
3ax¡4by+5cz

User-defined binary and relational operators

$34\mathrm{x}^2 a \# 13bc$
$34x^2 a \# 13bc$

Reference Guide

Description of Spacing Commands	
LaTeX Code	Description
\	space equal to the current font size(=18 mu)
\ ,	3/18 (=3 mu)
\ :	4/18 (=4 mu)
\ ;	5/18 (=5 mu)
\ ;	-3/18 (=-3 mu)
\ (space after backslash)	equivalent of space in normal text
\ *qquad*	twice of (=-36 mu)